Lecture 3:
SAP ERP

Business Applications (Betriebswirtschaftliche Anwendungen)

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SoSe 2026

Hochschule für Technik
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Recap last week

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Course content

1	Course Kickoff	7	AI agent ecosystem & Retro
2	Introduction to ERP systems	8	AI agents in business workflows
3	SAP ERP	9	Agentic ERP in S/4 HANA
4	Principles of ERP systems	10	OpenClaw & Test preparation
5	Modern ERP concepts	11	E-Test
6	Fundamentals of AI agents		

Learning objectives

At the end of the lecture, the students are able to …

- explain the **role of SAP** as a global ERP market leader
- describe the **core concepts and architecture of S/4HANA**
- navigate **SAP Fiori** as a modern user interface
- outline **migration paths** from legacy systems to S/4HANA

Agenda

1 **SAP SE**

2 S/4HANA Core Concepts

3 S/4HANA Architecture

4 SAP Fiori

5 Migration to S/4HANA

SAP profile

Market Leader
in Enterprise Software

#3
Largest Software Company Globally

110,000+
Employees from 157+ Countries

€36.8B
Total Revenue (FY2025)

€21B
Cloud Revenue (FY2025)

€77B
Cloud Backlog – grown by 30% in 2025

Sources: Global Company Information, 2025 SAP Integrated Report, SAP Announcements, Stockanalysis SAP, SAP UCC Magdeburg (2025)

The story of SAP (1/2)

1972 – SAP founded by 5 former IBM engineers in Mannheim

1981 – SAP R/2 launched for mainframe computing

1992 – SAP R/3 introduced 3-tier client-server architecture

1996 – R/3 becomes web-enabled for the internet era

2002 – SAP turns 30 as world's 3rd-largest software company

Sources: SAP UCC Magdeburg (2025)

The story of SAP (2/2)

2005
SAP ERP launched as successor to R/3

2009
Business Suite 7 released (better business processes, reduced costs)

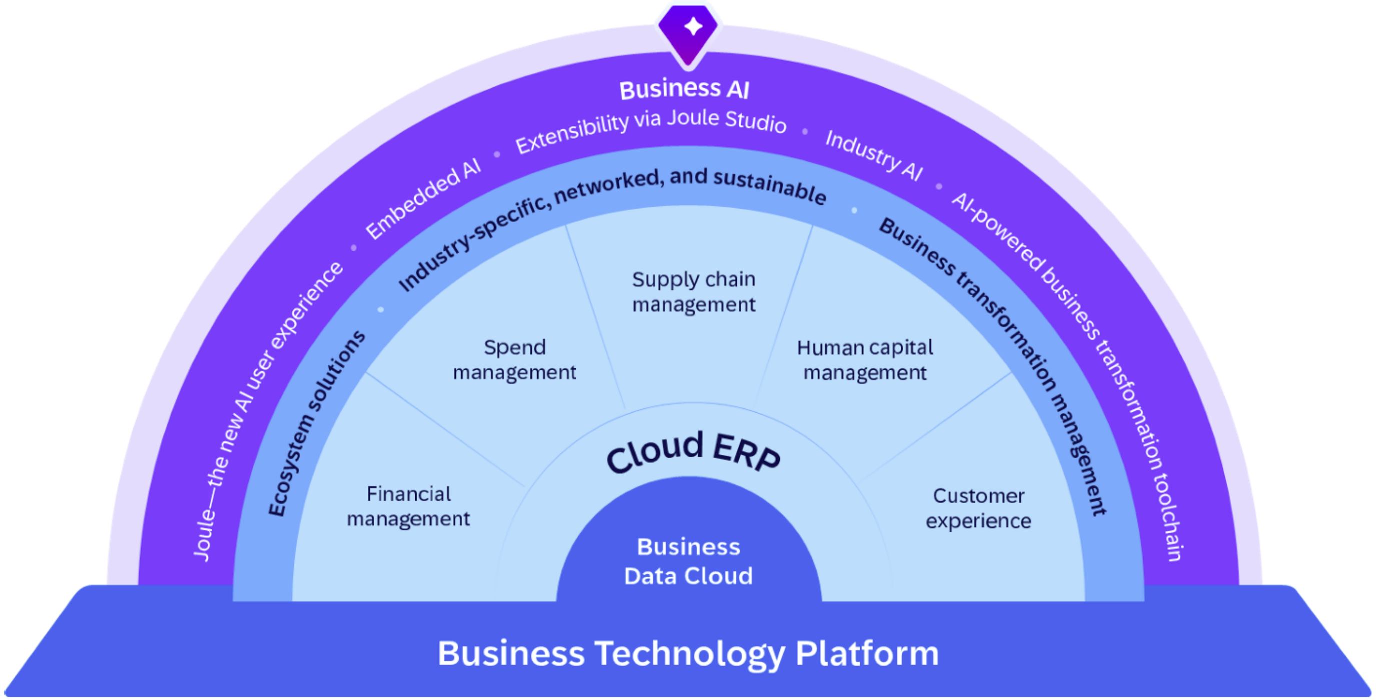
2011
SAP HANA in-memory platform introduced

2013
SAP Business Suite moves to SAP HANA – combining apps

2015
S/4HANA launched as next-gen ERP suite

Sources: SAP UCC Magdeburg (2025)

SAP Business Suite



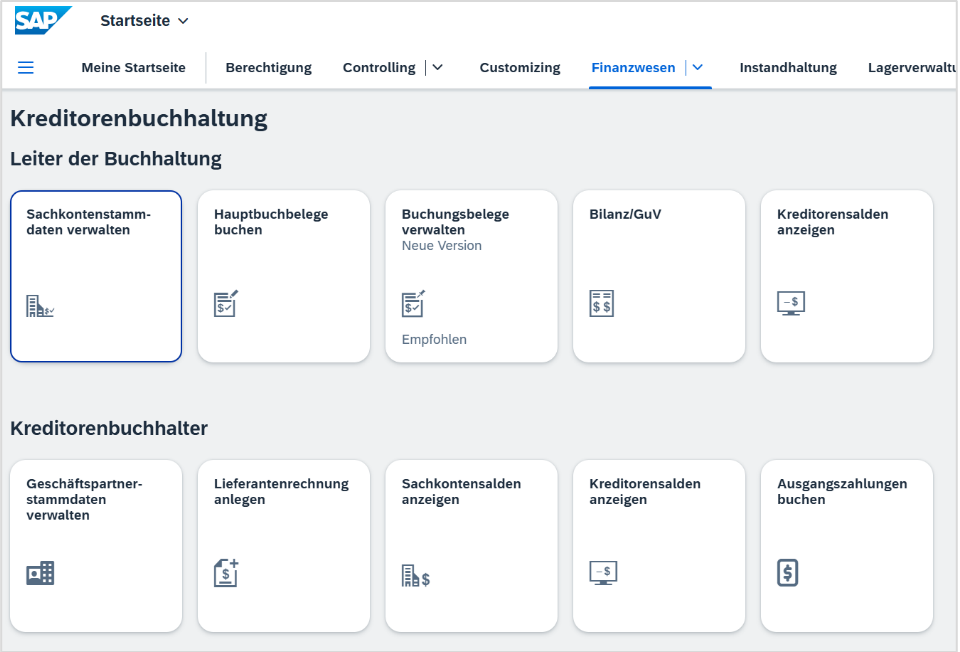
Sources: 2025 SAP Integrated Report

Agenda

1. SAP SE
2. **S/4HANA Core Concepts**
3. S/4HANA Architecture
4. SAP Fiori
5. Migration to S/4HANA

What is SAP S/4 HANA? (1/2)

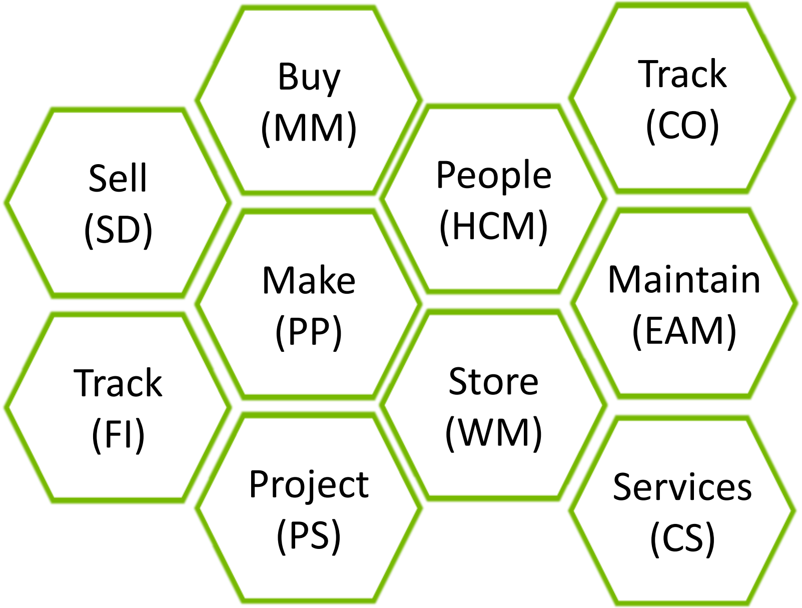
- Next-gen business suite
- Biggest innovation since SAP R/3
- Connects people, business networks and devices
- Operates with real-time data
- Centrally manages master data for partners, customers and suppliers



Sources: SAP UCC Magdeburg (2025)

What is SAP S/4 HANA? (2/2)

- Enables companies to conduct and optimize business processes
- Supports smooth operations within the organization
- Scalable and flexible environment
- Collections of logically related transactions for business functions



Sources: SAP UCC Magdeburg (2025)

Activity: What type of data do ERP systems process?

- Own brainstorming with researching (5 minutes)
- Exchanging ideas with seat neighbor (5 minutes)
- Plenum discussion (5 minutes)

Data types in ERP systems

Org. Data	Master Data	Transaction Data
Client	Person	Inquiry
Company Code	Material	Quotation
Plant	Customer	Purchase Order
Storage Location	Supplier	Invoice
…	…	…

Sources: SAP UCC Magdeburg (2025)

Organizational data

Client *(Mandant)*
Highest level – represents entire SAP system

Company Code *(Buchungskreis)*
Legal entity for financial reporting

Plant *(Werk)*
Production or logistics site

Storage Location *(Lagerort)*
Physical warehouse location

Cross-Cutting Units

Sales Organization *(Vertriebsorg.)*
Sales structure for distribution

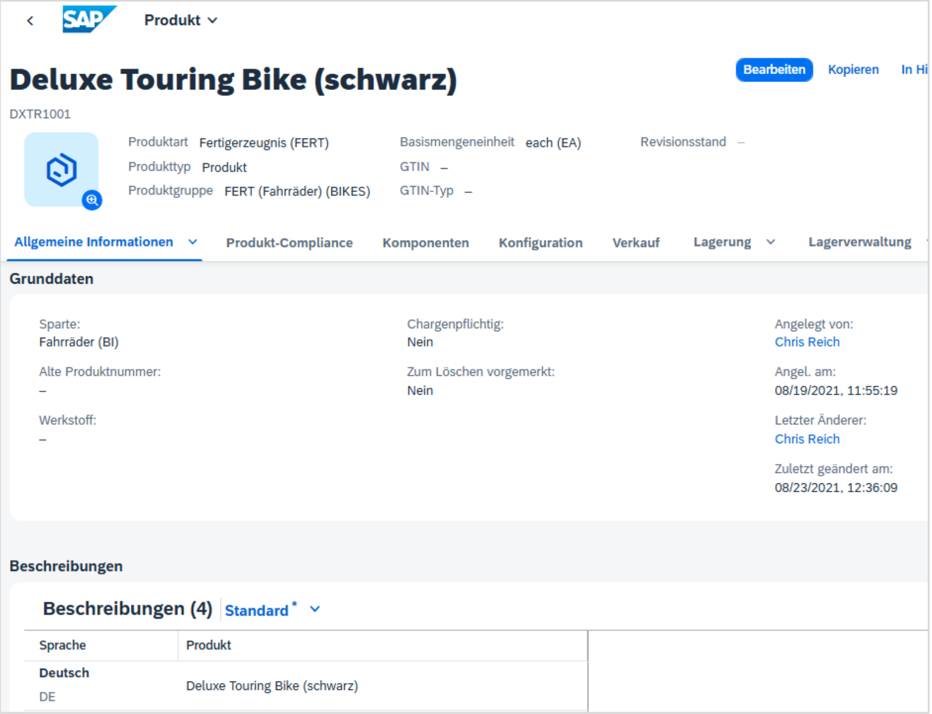
Division *(Sparte)*
Product line grouping

Business Area *(Geschäftsbereich)*
Internal reporting segment

SAP organizes enterprises into a hierarchy of organizational units. The client is the top-level container. Cross-cutting units like sales organization span multiple levels of the hierarchy.

Master data

- Long-term data
- Typically represent data records
- Example from Sales
  - Customer master data
  - Material master data
  - Condition master data



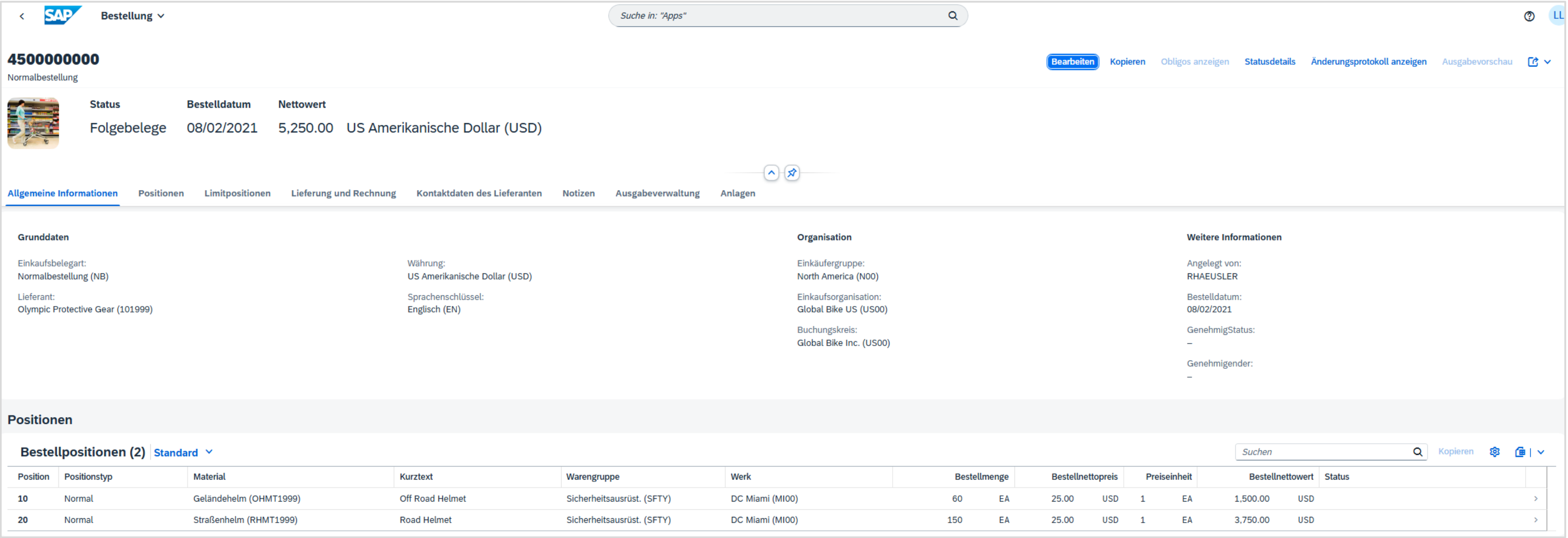
Sources: SAP UCC Magdeburg (2025)

Transaction data (1/2)

- Process-related data that is short-lived and linked to specific master data
- Contains organizational units, master data and situational data
- Example: Purchase order
  - **Organizational units:** Client, company code, sales organization
  - **Master data:** Customer master data, material master data
  - **Situational data:** Date, quantity, etc.

Sources: SAP UCC Magdeburg (2025)

Transaction data (2/2)



Sources: SAP UCC Magdeburg (2025)

Documents *(Belege)*

- Records generated when executing a transaction
- Represents a log of the business transaction
- Contains all relevant, predefined information from master data and organizational units
- Examples: Sales document, material document, accounting document
- Document flow
  - The document flow and order status allow you to determine the status of an order at any point in time
  - SAP updates the status each time a doc change occurs

Sources: SAP UCC Magdeburg (2025)

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3 **S/4HANA Architecture**

4 SAP Fiori

5 Migration to S/4HANA

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Digital transformation

Technology Drivers
IoT, Cloud Computing, Big Data, Mobility

R&D
Forecasting, simulations, new technologies

Business Changes
New org structures, agile processes, customer insights

Digital Transfor-mation

Customer Forces
Lifestyle dynamics, social media influence, info access

Digital transformation is driven by converging forces. SAP S/4HANA is designed to enable this transformation across all dimensions.

Sources: SAP UCC Magdeburg (2025)

SAP S/4 HANA architecture

SAP Fiori

User Experience Layer

SAP S/4HANA Applications

Finance | Logistics | Manufacturing | Sales | HR | Procurement | ...

SAP HANA

In-Memory Database | Analytics | Real-Time Processing

Infrastructure

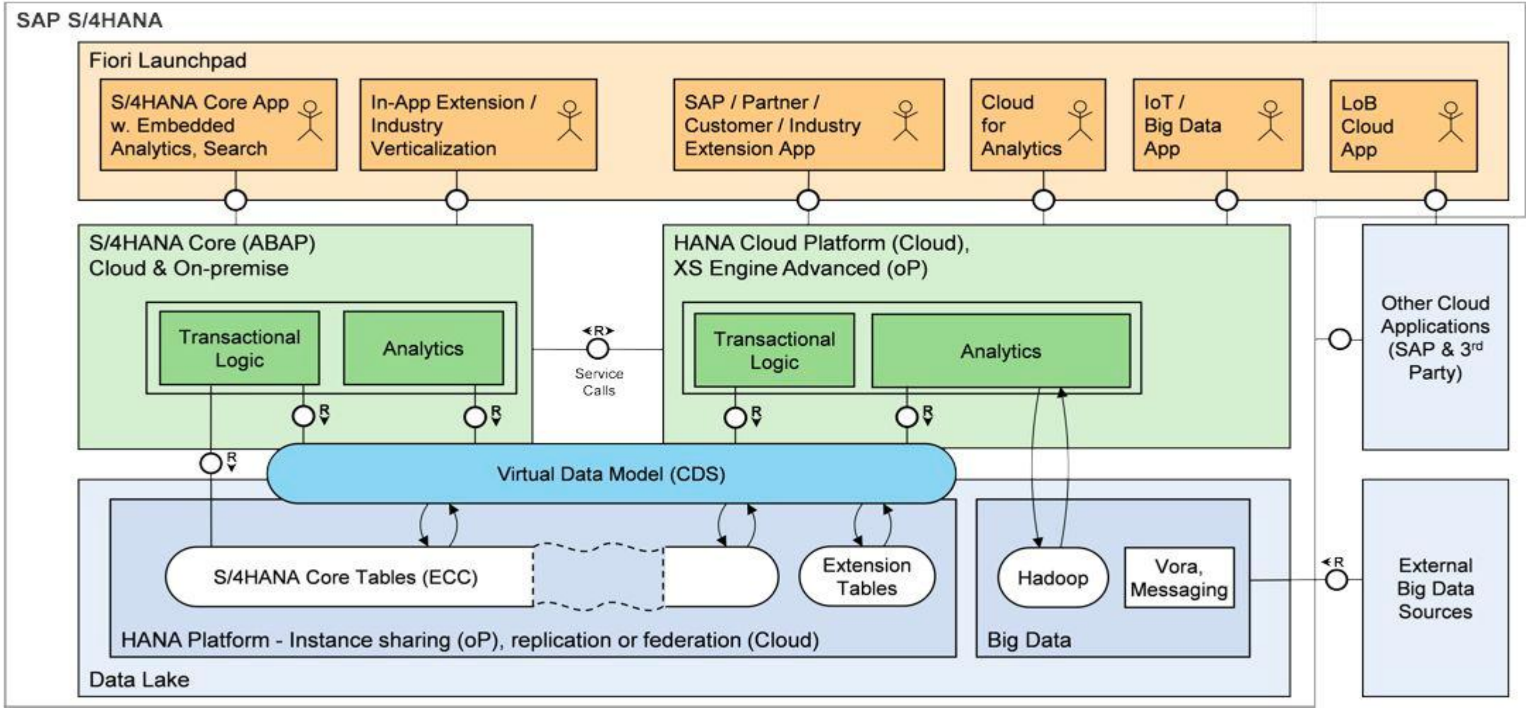
On-Premise | Private Cloud | Public Cloud (Hyperscalers)

SAP Business Technology Platform

Extensions | Integration | AI

Sources: SAP SE product documentation

SAP S/4 HANA architecture



Sources: SAP UCC Magdeburg (2025)

SAP S/4 HANA On-Premise vs. Cloud infrastructure

On-Premise	Cloud
Traditional licensing model	Subscription licensing model
Customer controls deployment and maintenance	SAP deploys and provides maintenance
Hardware located at company site	Hardware located in SAP ecosystem
Full control over private data	Automatic quarterly upgrades
Longer release cycles	Current release cycles
Custom requirements implementable	In-app extensibility (with limited ABAP)

Sources: SAP UCC Magdeburg (2025)

Three predominant SAP S/4 HANA features

In-Memory Database

Runs on SAP HANA

- ✓ Column-oriented DB
- ✓ In-memory processing
- ✓ Data compression
- ✓ Parallel processing

New Technology

OLAP + OLTP combined

- ✓ Real-time analytics
- ✓ Compact forecasts
- ✓ Embedded analytics
- ✓ Simplified data model

Modern Design

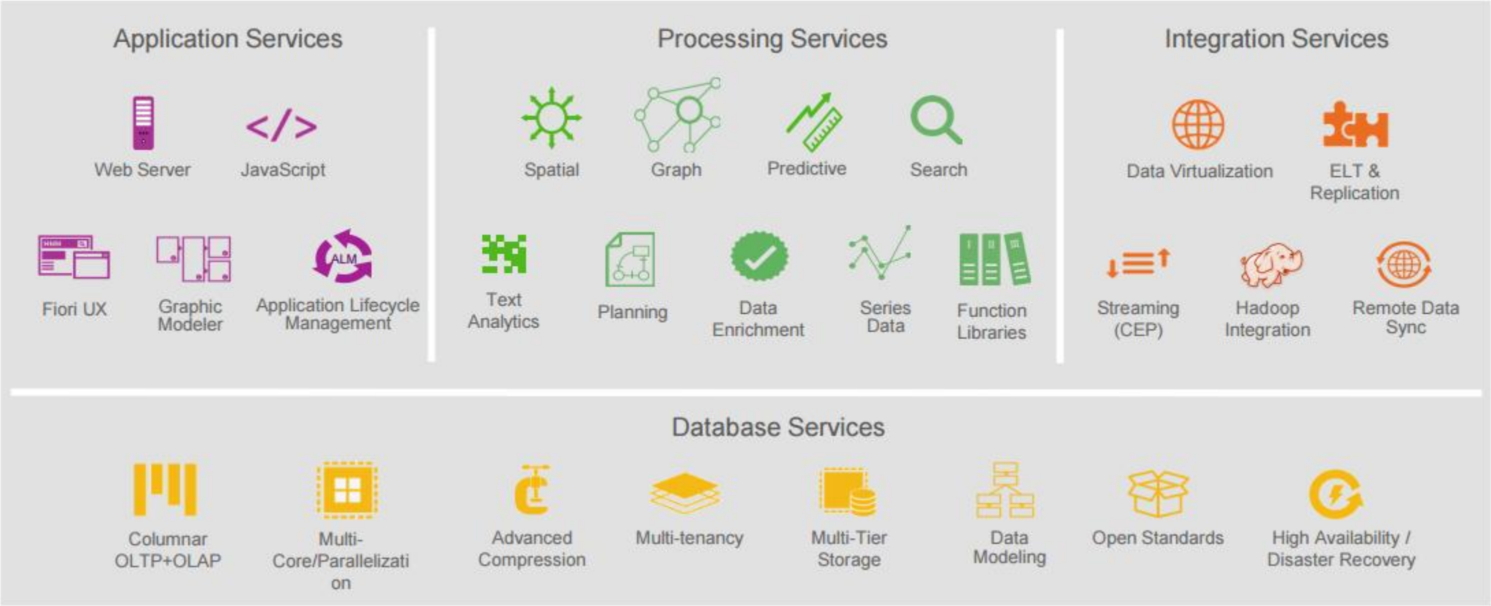
SAP Fiori UX

- ✓ Hardware independent
- ✓ Mobile device support
- ✓ Intuitive user interface
- ✓ Role-based applications

Sources: SAP UCC Magdeburg (2025)

Deep-Dive: SAP HANA

In-memory database for fast data processing



Sources: SAP UCC Magdeburg (2025)

Leisure time

What content from this lecture can you recall so far?

Reflect together with the person sitting next to you

5 minutes

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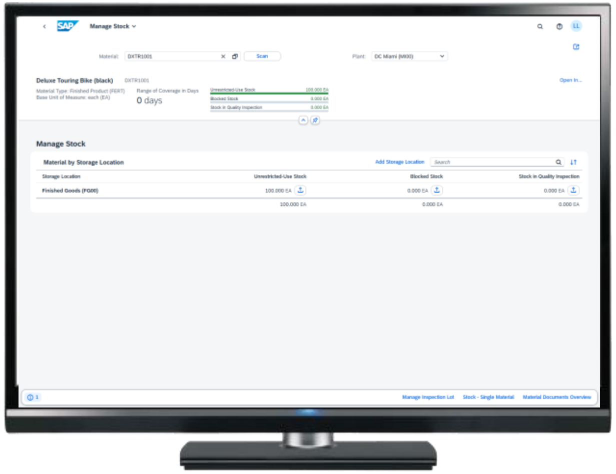
5. Migration to S/4HANA

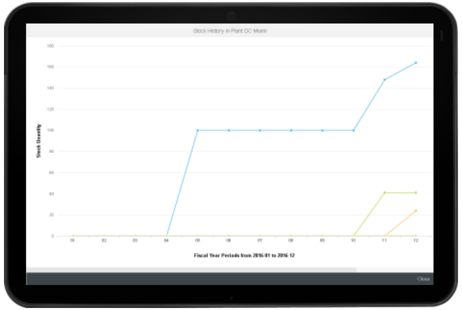
SAP Fiori supports different devices

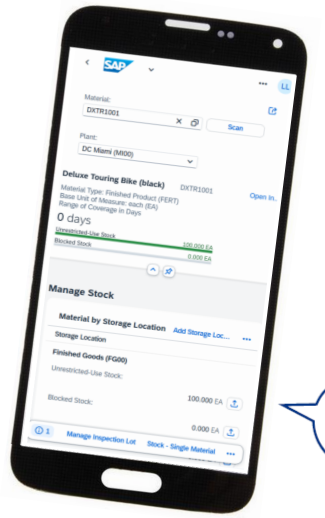
SAP S/4HANA uses the SAP Fiori user interface



SAP S/4HANA

- Hardware independent
- Real-time

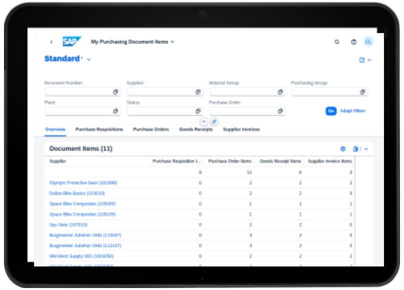


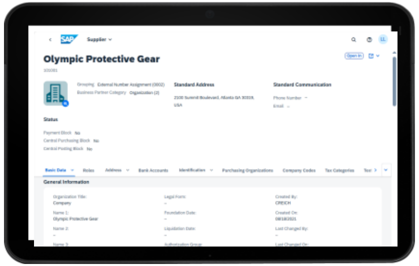


Sources: SAP UCC Magdeburg (2025)

SAP Fiori uses different types of apps

- **Transactional Apps**
  Access to transactions such as Create, Change or Display with guided navigation
- **Analytical Apps**
  Visual and graphical overview of business data
- **Fact Sheet Apps**
  Essential information about objects and navigation between related apps

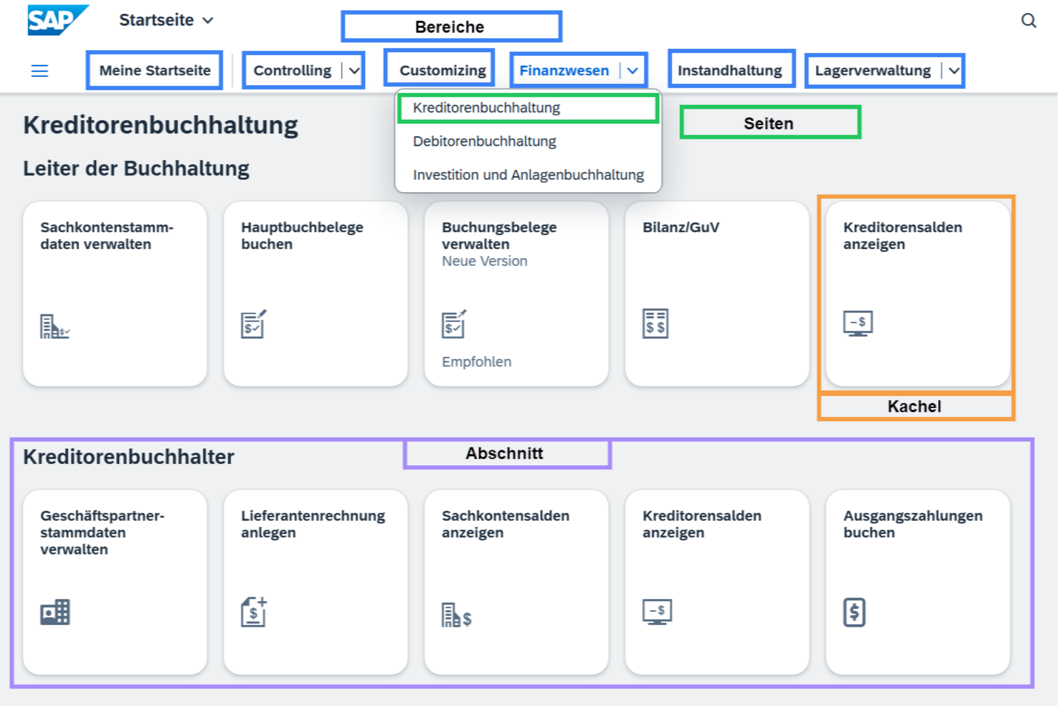




Sources: SAP UCC Magdeburg (2025)

SAP Fiori Launchpad navigation

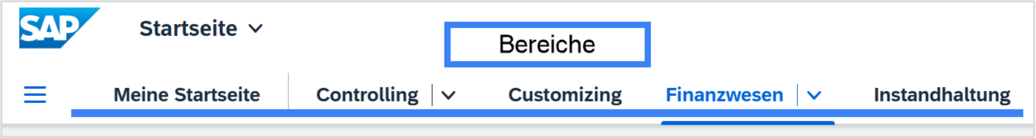
- Modern user interface
- Unified, role-based entry point for applications
- Structure: **Spaces** → **Pages** → **Sections** → **Tiles (Apps)**
- Designed for an intuitive and user-friendly experience
- Default user interface in **SAP S/4HANA**



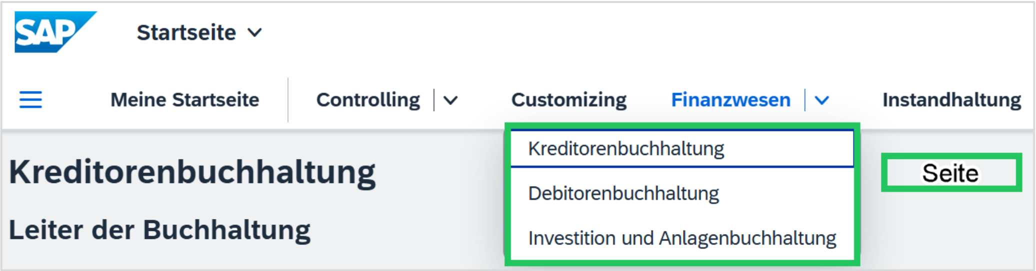
Sources: SAP UCC Magdeburg (2025)

Deep-Dive: SAP Fiori Launchpad (1/2)

- **Space**
  Groups thematically related applications for a specific user role or business area



- **Page**
  The main workspace where applications and information are displayed

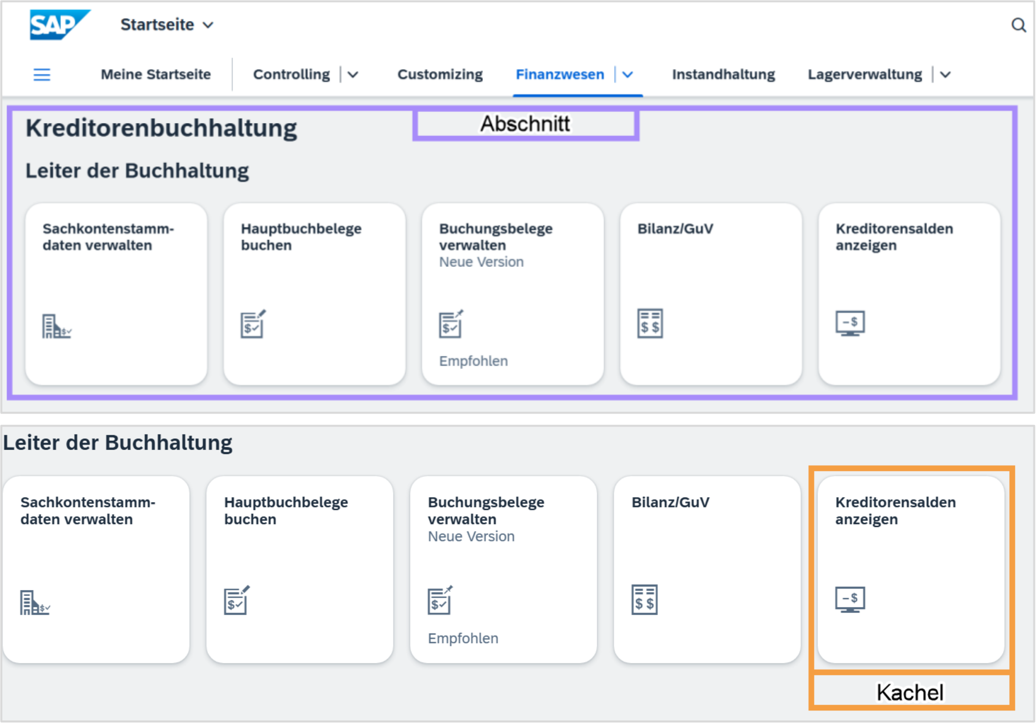


Sources: SAP UCC Magdeburg (2025)

Deep-Dive: SAP Fiori Launchpad (2/2)

- **Section**
  Used to further subdivide pages and group tiles by specific tasks or processes

- **Tiles**
  Direct entry point to an application – the smallest interactive elements

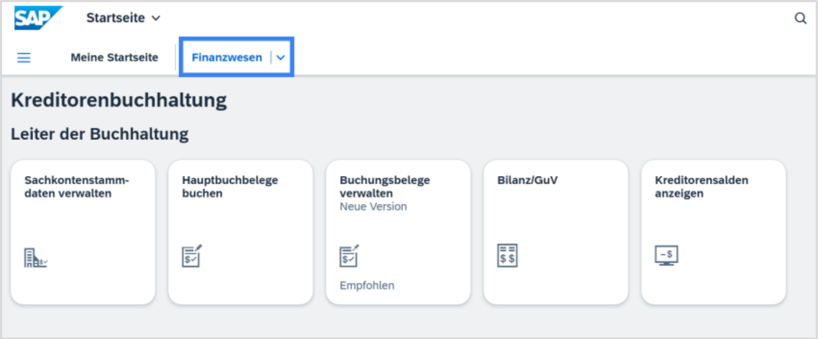


Sources: SAP UCC Magdeburg (2025)

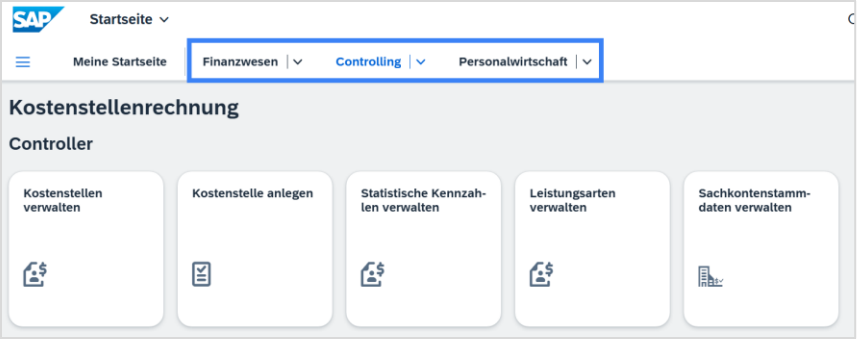
Business roles

- Foundation of the authorization structure
- The assigned role determines which content a user can see and access

Authorization for one role



Authorization for multiple roles



Sources: SAP UCC Magdeburg (2025)

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3. S/4HANA Architecture
4. SAP Fiori
5. **Migration to S/4HANA**

Three migration paths to SAP S/4HANA

1 New Implementation

Greenfield Approach

Fresh SAP S/4HANA installation for customers migrating from legacy or non-SAP systems

SAP S/4HANA On-Premise

SAP S/4HANA Cloud Edition

2 System Conversion

Brownfield Approach

In-place conversion of an existing SAP ERP system to SAP S/4HANA

SAP S/4HANA On-Premise

SAP S/4HANA

3 Landscape Transformation

Selective Approach

Consolidation or selective transformation of an existing system landscape

SAP S/4HANA On-Premise

SAP S/4HANA Cloud Edition

Sources: SAP UCC Magdeburg (2025)

1 New implementation

Scenario

Fresh SAP S/4HANA installation for customers migrating from legacy or non-SAP systems

Benefits

- Restructuring and simplification based on ready-to-run business processes
- Migration of predefined objects, guided config
- Reduced transformation time and costs
- Rapid adoption of new innovations

Project Duration

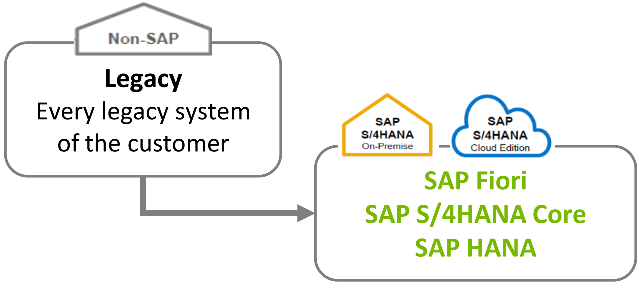
- Migration data volume (e.g. materials, customers)
- Complexity of data migration

Approach

1. Install SAP S/4HANA
2. Load source data from legacy system

Tools

- SAP Data Services (DS; on-premise)
- SAP Landscape Transformation (LT; cloud)



Sources: SAP UCC Magdeburg (2025)

2 System conversion

Scenario

- In-place conversion of existing SAP ERP to SAP S/4HANA
- Migration of DB, SAP NetWeaver and applications

Benefits

- Migration without re-implementation
- No disruption to ongoing business operations
- Reassessment of custom modifications and existing workflows

Project Duration

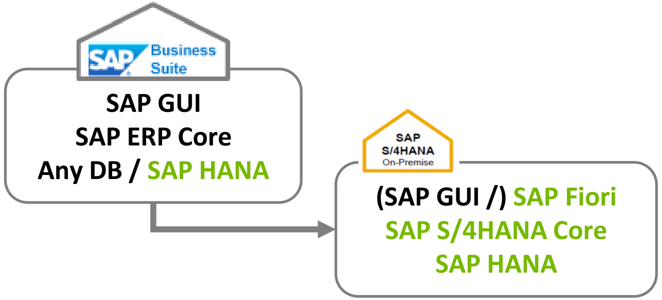
- Technical: Number of systems and database size
- Functional: Number of company codes, books, business requirements

Approach

1. Review add-ons and solutions for compatibility
2. Verify components and prerequisites
3. Start the conversion

Tools

Software Update Manager (SUM)



Sources: SAP UCC Magdeburg (2025)

3 Landscape transformation

Scenario

Consolidation or selective transformation of existing landscape with SAP S/4HANA

Benefits

- Selective data transformation enables a phased approach with high ROI and low TCO
- Landscape consolidation with harmonized processes and unified master data lowers TCO

Project Duration

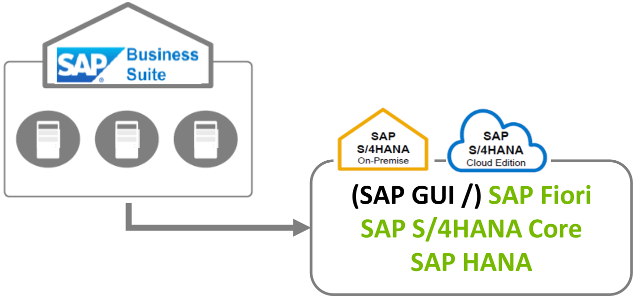
- General: Depends on sub-scenario (e.g., system consolidation, selective transformation)
- Specific: Number of systems to consolidate and volume of selected data

Approach

1. Consolidation: Merge clients from different source systems to SAP S/4HANA using SAP LT
2. Selective data transformation: Migrate selected SAP applications (e.g., central finance)

Tools

SAP Landscape Transformation (LT)



Sources: SAP UCC Magdeburg (2025)

Thank you for your attention

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